

kinds of social behavior, and sustain communications in a natural language, allowing for a wider reach. Machine learning algorithms are now as good as humans at writing these messages, and can come up with them faster, and without tiring. This allows phishing attempts to leave the realm of the inbox, and say, use a chatbot instead.

Targeted phishing attacks, known as spear phishing, have become more focused over the last year. A report published by Ironscales, an automated phishing prevention provider, showed that over 77 percent of the attacks targeted 10 inboxes or less. 33 percent of the spear phishing attacks were directed towards just one individual. These kind of emails slip past the traditional spam filters in place.

For high value targets, instead of sending out emails en masse, the attacker might choose to target the emails on an individual basis. Information farmed from social networks can be used to draw up profiles of an individual, allowing a cybercriminal to figure out what kind of approach is likely to work best. The spear phishing attacks, can be based around what motivates an individual, or scares them the most. These kind of phishing attacks are more likely to focus on gaining access to the communications and documents of the target, as against details of their bank accounts. According to Phishlabs, 95 percent of espionage attacks involve phishing, and 20 percent is the average spear phishing success rate. Ministers, high ranking officials, and those in the top tier of management are most likely to be targeted. Academics, activists and journalists are other high value targets.

To stay safe from spoofing emails, one of the simplest precautions one can take is to make a call to the person involved, to verify the need for the requested action, such as a transfer of funds. While keying in login information, be sure to verify that the website is using an https protocol, and that it is the actual website by carefully scrutinising the url. Enable two factor authentication wherever possible. There is no legitimate reason to provide your personal login details to any person, for any reason whatsoever. ■

RANSOMWARE THAT MAKES YOU WANNACRY

Ransomware was one of the biggest cybersecurity concerns over the course of 2017. Ransomware as a service is a business model in darknet marketplaces, which allows users to buy ransomware attacks. Ransomware developers stand to earn more than double the salary they would have got while working for a regular company. Three of the biggest attacks, Petya, WannaCry and Bad Rabbit, all used exploits from a leaked NSA cache. Patches for the exploits, EternalBlue and EternalRomance had already been released at the time of the attacks. Technically, the ransomware did not even use zerodays, or previously undiscovered security flaws.

Ransomware attacks are only expected to increase in 2018, and get more creative as well. The attackers have now understood which kind of organi-

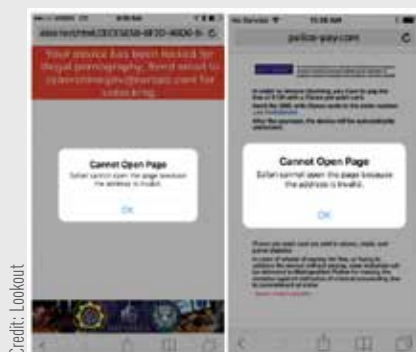
sations are vulnerable to attacks, and which kind of organisations have strong defences. For example, a logistics company or a health services company are both likely to have fewer defences in place, rather than say a company that offers financial services. Those looking to actually earn the money from a ransomware attack, are likely to target a few organisations instead of lots of individuals. Research has shown that companies are more than willing to pay up the amounts

that are demanded during ransomware attacks. That does not mean that an attack on companies will not adversely affect individuals.

One of the high value targets for ransomware attacks are cloud services companies. If storing data in the cloud, it is best to opt for one of the larger companies which have adequate resources to hire the best cybersecurity professionals, such as Amazon or Google.

So far ransomware attacks have only been restricted to Windows machines. In 2018, expect ransomware attacks to spread to Android/iOS devices. In fact, smartphone users have been fooled by malware that shows messages similar to a ransomware, demands a payment, and does not really even lock up data on the device. Another lucrative target for ransomware attacks in 2018, are a whole bunch of new devices that are known to have especially lax security. Expect IoT ransomware to lock up devices, with the price point set below the cost of just buying a new device.

The best way to protect yourself against ransomware attacks is to install apps from trusted marketplaces, not click on suspicious links in emails, apply the latest security patches, and to store all critical data in an air gapped hard disk. If an attack is successful on any of your devices, never pay the ransom, as making the payment is no guarantee that the data will in fact, be decrypted. ■



Credit: Lookout

A fake ransomware or scareware attack on an iOS device

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